

EXP 5

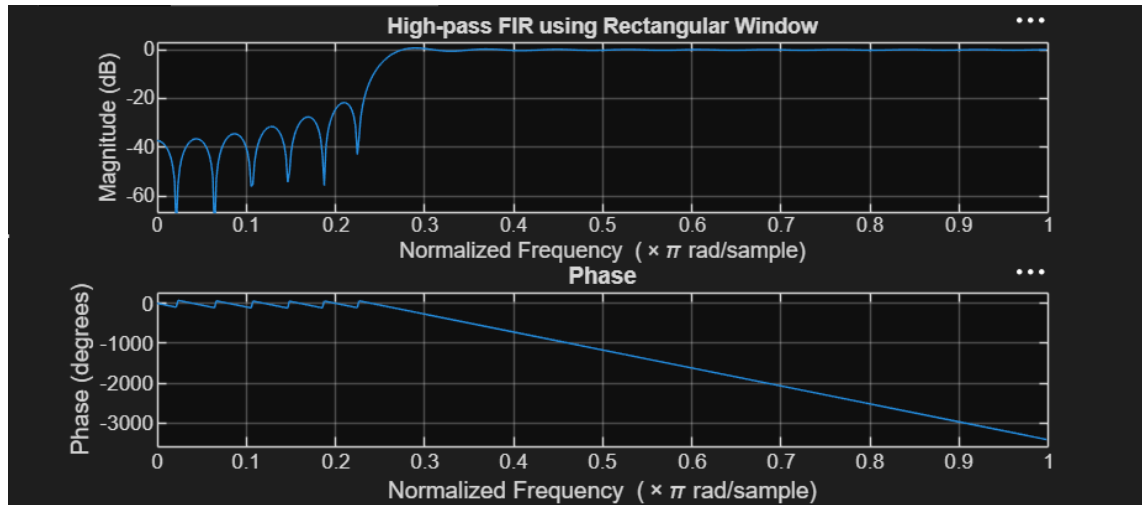
CODE:

```
clc; clear; close all;
%%% --- Input Parameters ---
fp = 1000; % High-pass cutoff frequency (Hz)
fs = 8000; % Sampling frequency (Hz)
N = 50; % Filter order
wc = fp/(fs/2); % Normalized cutoff (0–1)
%%% --- 1. FIR High Pass using Rectangular Window ---
b_rect = fir1(N, wc, 'high', rectwin(N+1));
disp('--- High-Pass Coefficients (Rectangular Window) ---');
disp(b_rect');
%%% --- 2. FIR High Pass using Hamming Window ---
b_hamm = fir1(N, wc, 'high', hamming(N+1));
disp('--- High-Pass Coefficients (Hamming Window) ---');
disp(b_hamm');
%%% --- 3. FIR High Pass using Hanning Window ---
b_hann = fir1(N, wc, 'high', hanning(N+1));
disp('--- High-Pass Coefficients (Hanning Window) ---');
disp(b_hann');
%%% --- Frequency Response Plots ---
figure;
freqz(b_rect,1);
title('High-pass FIR using Rectangular Window');
figure;
freqz(b_hamm,1);
title('High-pass FIR using Hamming Window');
figure;
freqz(b_hann,1);
title('High-pass FIR using Hanning Window');
```

OUTPUT:

```
--- High-Pass Coefficients (Rectangular Window) ---
-0.0089
0
0.0097
0.0143
0.0106
0
-0.0117
-0.0175
```

-0.0131
0
0.0148
0.0225
0.0171
0
-0.0202
-0.0314
-0.0247
0
0.0318
0.0524
0.0445
0
-0.0741
-0.1572
-0.2223
0.7407
-0.2223
-0.1572
-0.0741
0
0.0445
0.0524
0.0318
0
-0.0247
-0.0314
-0.0202
0
0.0171
0.0225
0.0148
0
-0.0131
-0.0175
-0.0117
0
0.0106
0.0143
0.0097
0
-0.0089



EXP 5.2

OUTPUT:

--- High-Pass Coefficients (Hamming Window) ---

-0.0007

0

0.0009

0.0016

0.0015

0

-0.0024

-0.0044

-0.0039

0

0.0060

0.0103

0.0088

0

-0.0128

-0.0217

-0.0184

0

0.0268

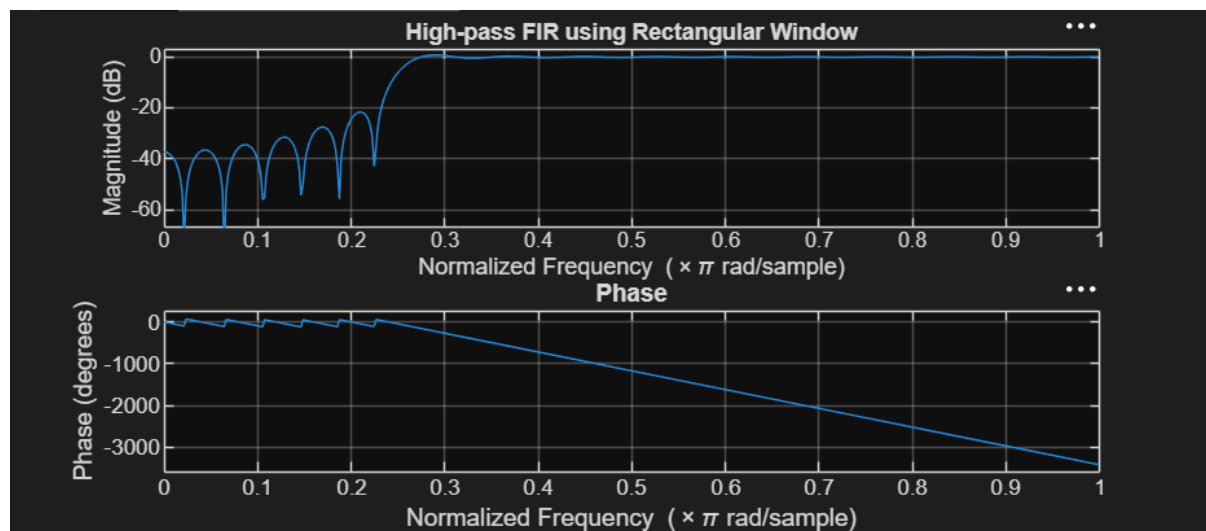
0.0464

0.0410

0

-0.0725

-0.1567
-0.2240
0.7493
-0.2240
-0.1567
-0.0725
0
0.0410
0.0464
0.0268
0
-0.0184
-0.0217
-0.0128
0
0.0088
0.0103
0.0060
0
-0.0039
-0.0044
-0.0024
0
0.0015
0.0016
0.0009
0
-0.0007



EXP 5.3

OUTPUT:

--- High-Pass Coefficients (Hanning Window) ---

-0.0000

0

0.0003

0.0008

0.0009

0

-0.0020

-0.0038

-0.0035

0

0.0057

0.0100

0.0087

0

-0.0127

-0.0216

-0.0183

0

0.0267

0.0464

0.0410

0

-0.0726

-0.1568

-0.2243

0.7500

-0.2243

-0.1568

-0.0726

0

0.0410

0.0464

0.0267

0

-0.0183

-0.0216

-0.0127

0

0.0087

0.0100
0.0057
0
-0.0035
-0.0038
-0.0020
0
0.0009
0.0008
0.0003
0
-0.0000

